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Experiment 7

Aim: Write a program to simulate Load balancing algorithm.

Theory:

- Load balancing ensures even distribution of workloads across multiple servers or nodes in a distributed system.
- This prevents overloading of any single node, improving performance, fault tolerance and scalability.
- Load balancing algorithms can be classified into static and dynamic approaches:

1) Static Load balancing
⇒ Assigns tasks before execution based on predefined rules.
- Works well for predictable and stable workloads.
- Example: Round Robin, Randomized algorithm

2) Dynamic Load balancing
⇒ Adjusts resource allocation during execution based on system load.
- Requires real-time monitoring of system performance.
- For e.g., Least connection, Weighted Round Robin.

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- Key characterstics :

1) Traffic distribution : To keep any one server from becoming overburdened, load balancers divide incoming requests evenly among several servers.
2) High Availability : Application's reliability and availability are improved by load balancers, which divide traffic among several servers
3) Scalability : By making it simple to add servers or resources to meet growing traffic demands, load balancers enable horizontal scaling.
4) Optimization : Load balancers optimize resource utilization, ensuring efficient use of servers capacity and preventing bottlenecks.
5) Health monitoring : Load balancers often monitor the health of servers, directing traffic away from servers experiencing issues or downtime.
6) SSL Termination : Some load balancers can handle SSL/TLS encryption and decryption, offloading this resource-intesive task from servers.

(A) Pradya